Assignment 8: Multimodal Architecture using Graph Neural Networks

All experiments use 5,000 products from the Fashion Product Images (Small) dataset, classified into 5 master categories (Apparel, Accessories, Footwear, Free Items, Personal Care). Image embeddings (64-d) are extracted from a pretrained ResNet18 and text embeddings (64-d) via SVD-projected bag-of-words on product names. The data is split 80/20 for training and testing.

1 Part A: GNN on Bipartite Graphs

We construct a bipartite graph with two node types: image nodes and text nodes (5,000 each, 10,000 total). Three edge types connect them: Image-Image and Text-Text edges via k-NN (k=5) on cosine similarity (25,000 edges each), and Image-Text edges linking each product's image and text nodes bidirectionally (10,000 edges). A single message passing layer aggregates neighbour features ($\mathbf{m}_i = \sum_j W_{\text{msg}}\mathbf{h}_j$, update: $\mathbf{h}'_i = \text{ReLU}(W_{\text{upd}}[\mathbf{h}_i \| \mathbf{m}_i])$). Product-level predictions are obtained by averaging each product's image and text node representations.

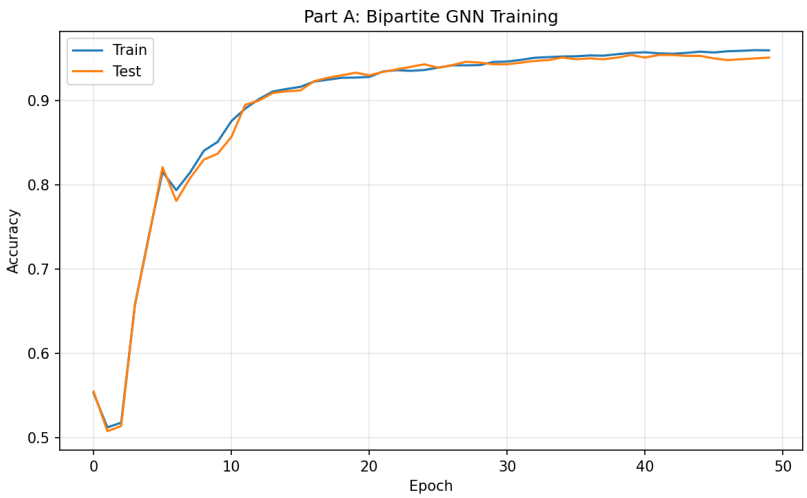

Figure 1: Part A: Bipartite GNN training and test accuracy.

The bipartite GNN reaches 95.4% test accuracy. The cross-modal edges are crucial: they allow image features to inform text representations and vice versa during message passing, creating richer multimodal node embeddings than either modality alone.

2 Part B: Fusion Strategies

We compare the bipartite GNN against two fusion approaches and implement cross-modal retrieval.

Early fusion concatenates image and text embeddings into a 128-d vector, projects through an MLP to 64-d, then runs one message passing layer on a k-NN graph built from the fused features. **Late fusion** runs two independent GNNs on separate image and text similarity graphs, then concatenates their outputs and classifies via an MLP.

Model	Best Test Acc
Bipartite GNN (Part A)	0.954
Early Fusion GNN	**0.981**
Late Fusion GNN	0.977

Table 1: Fusion strategy comparison.

Early fusion outperforms both alternatives because fusing features before graph construction creates a unified similarity space where neighbours are truly multimodally similar, leading to more informative message passing. Late fusion performs well but lacks cross-modal interaction during message passing – each GNN only sees its own modality. The bipartite GNN falls between, as cross-modal edges exist but the graph is more complex to optimise.

Cross-modal retrieval uses the early fusion GNN: a query from one modality (with zeros for the other) is projected through the model's fusion layer and compared via cosine similarity against all GNN-learned embeddings. Retrieval accuracy is low because zeroing out an entire modality creates an out-of-distribution input that the fusion MLP was not trained to handle – a known limitation of simple concatenation-based fusion.

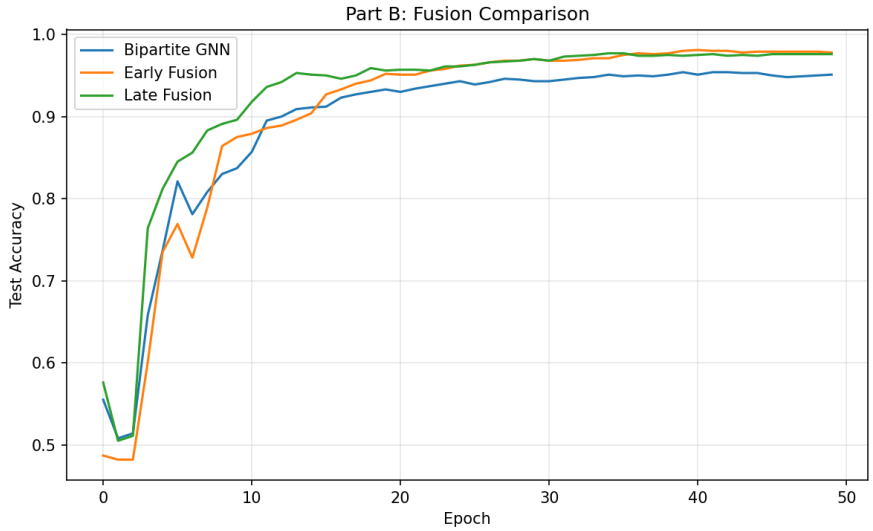

Figure 2: Test accuracy comparison across fusion strategies.

3 Part C: Semi-supervised Learning

We train with limited labels (5%, 10%, 20%) using four strategies: supervised-only cross-entropy, pseudo-labelling (high-confidence predictions used as labels every 5 epochs), Mean-Teacher (EMA teacher with consistency regularisation), and both combined.

Method	5%	10%	20%
Supervised only	0.933	0.942	0.955
+ Pseudo-labels	0.925	0.944	0.955
+ Mean-Teacher	0.932	**0.953**	0.953
+ Both	**0.936**	0.947	**0.958**

Table 2: Semi-supervised learning across label fractions.

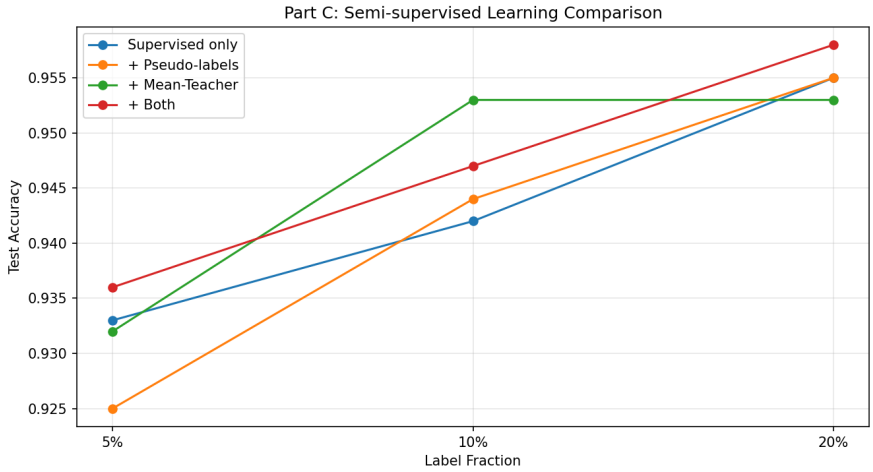

Figure 3: Test accuracy vs label fraction for each semi-supervised method.

The combined approach (pseudo-labels + Mean-Teacher) performs best overall, gaining 0.3–0.5 points over supervised-only at low label fractions. Mean-Teacher is particularly effective at 10% labels (+1.1 points) because consistency regularisation leverages the GNN's graph structure to propagate information from labelled to unlabelled nodes through their shared neighbourhood. Pseudo-labels alone can hurt at very low label fractions (5%) because early pseudo-labels are unreliable and introduce noise.

4 Part D: Self-supervised Contrastive Learning

We pretrain a GNN encoder using InfoNCE contrastive loss on two augmented graph views created by edge dropout (20%), feature masking (20%), and feature jitter (σ=0.05). After 100 pretraining epochs, we freeze the encoder and evaluate with a linear probe and k-NN.

Evaluation	Accuracy
Linear probe (frozen)	0.721
k-NN (k=3)	0.911
k-NN (k=5)	0.912
k-NN (k=10)	0.885

Table 3: Contrastive embedding quality evaluation.

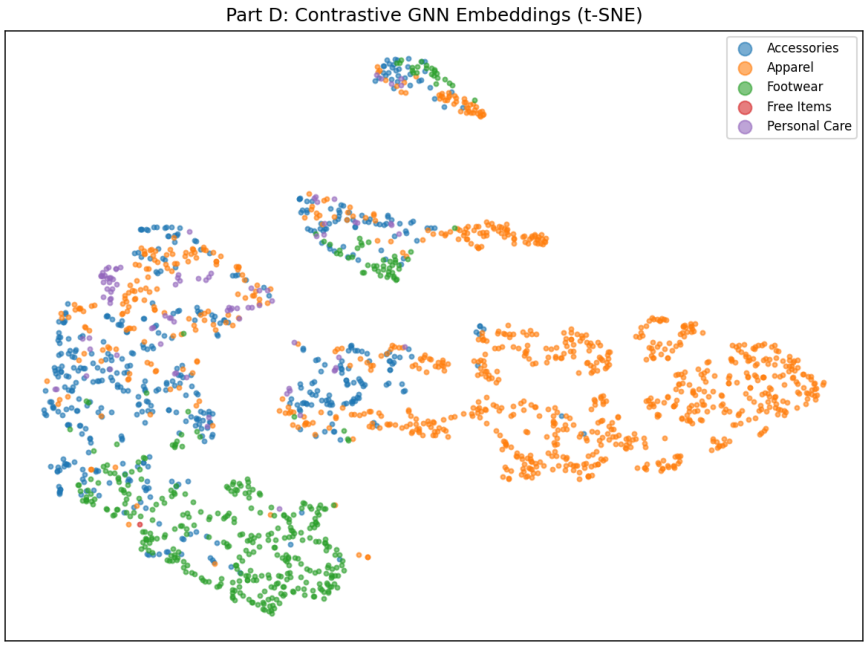

Figure 4: t-SNE of contrastive GNN embeddings coloured by category.

The k-NN accuracy (91.2%) significantly exceeds the linear probe (72.1%), indicating the learned embeddings form well-separated clusters in a non-linearly structured space. The t-SNE visualisation confirms clear category clusters with Footwear and Apparel well-separated. The gap to supervised accuracy (95–98%) is expected since contrastive pretraining uses no labels. These embeddings could serve as strong initialisations for downstream fine-tuning with limited labels, complementing the semi-supervised approaches from Part C.